School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Stochastic Processes: Part 1
Strict Sense Stationary (SSS), Wide Sense Stationary (WSS) and Ergodic Process

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Overview

This lecture covers basics of stochastic processes, statistical moments, stationary processes (SSS and WSS), and ergodic processes. It includes definitions, properties, derivations, proofs, and worked examples.

Definitions and Notation

Stochastic Process Definition

A stochastic process $X(\zeta, t)$ is an entire collection of waveforms or functions over time dependent on the outcome of a random experiment.

- The random output heavily depends on the experiment.
- Example: Tossing a coin gives heads or tails resulting in specific deterministic waveforms.

Realization

A specific measured waveform from the experiment is a **Realization** of the process.

Random Variable Representation

Fixing time $t = t_1$:

$$X(\zeta, t_1) = X(t_1)$$

Multiple time instants:

$$(X(t_1), X(t_2))$$

Probability Density Functions (PDF)

1st Order Characterization

$$t_1 \to f_X(x, t_1) \to F_X(x, t_1)$$

2nd Order Characterization

$$(t_1, t_2) \to f_{X_1 X_2}(x_1, x_2, t_1, t_2)$$

nth Order Characterization

$$(t_1, ..., t_n) \to f_{X_1,...,X_n}(x_1, ..., x_n, t_1, ..., t_n)$$

Mean (First Order Moment)

$$\mu_X(t) = E[X(t)]$$

$$\mu_X(t) = \int_{-\infty}^{\infty} x \cdot f_X(x, t)\, dx$$

Autocorrelation Function

$$R_{XX}(t_1, t_2) = E[X(t_1)X^*(t_2)]$$

$$R_{XX}(t_1, t_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1 x_2 f_{X_1 X_2}(x_1, x_2, t_1, t_2)\, dx_1 dx_2$$

Properties of Autocorrelation

Conjugate Symmetry

$$R_{XX}(t_1, t_2) = R^*_{XX}(t_2, t_1)$$

Proof Sketch

$$\begin{aligned} R^*_{XX}(t_2, t_1) &= (E[X(t_2)X^*(t_1)])^* \\ &= E[X^*(t_2)X(t_1)] = R_{XX}(t_1, t_2) \end{aligned}$$

Positive Definiteness

Autocorrelation forms a positive definite matrix.

Covariance Function

$$\begin{aligned} C_{XX}(t_1, t_2) &= E[(X(t_1) - \mu_X(t_1))(X(t_2) - \mu_X(t_2))] \\ &= R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2) \end{aligned}$$

Cross-Correlation

$$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$$

Cross-Covariance

$$C_{XY}(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(Y(t_2) - \mu_Y(t_2))]$$

Main Results

Strict Sense Stationary (SSS) Process

1st Order

$$f_X(x, t) = f_X(x, t + \tau) = g(x)$$

$$\mu_X(t) = \int xg(x)dx = \mu_X = \text{constant}$$

2nd Order

$$f_X(x_1, x_2; t_1, t_2) = f_X(x_1, x_2; t_1 + \tau, t_2 + \tau)$$

Choose $\tau = -t_2$:

$$f_X(x_1, x_2; t_1, t_2) = f_X(x_1, x_2; t_1 - t_2, 0)$$

$$R_{XX}(t_1, t_2) = \phi(t_1 - t_2)$$

$$R_{XX}(t_1, t_2) = R_{XX}(t_1 - t_2)$$

Wide Sense Stationary (WSS) Process

Condition 1: Constant Mean

$$E[X(t)] = \mu_X = \text{constant}, \quad \forall t$$

Condition 2: Autocorrelation depends only on lag

$$E[X(t_1)X^*(t_2)] = R_{XX}(\tau), \quad \tau = t_1 - t_2$$

WSS vs SSS

$$SSS \Rightarrow WSS \quad \text{but} \quad WSS \not\Rightarrow SSS$$

Gaussian Process Exception

$$WSS \Rightarrow SSS$$

Gaussian Process Definition

$$f_X(x) = \frac{1}{(2\pi)^{n/2}|R|^{1/2}} \exp\left(-\frac{1}{2}x^T R^{-1} x\right)$$

$$R = E[XX^T], \quad R_{ij} = E[X(t_i)X^*(t_j)]$$

Derivations / Proofs

Positive Definite Matrix Example

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

$$|A_1| = 2 > 0$$

$$|A_2| = 4 - 1 = 3 > 0$$

$$|A_3| = 4 > 0$$

Worked Examples

Example 1

$$X(t) = A\cos(\lambda t) + B\sin(\lambda t)$$

$$E[X(t)] = 0$$

$$R_{XX}(t_1, t_2) = E[A^2]\cos\lambda(t_1 - t_2)$$

$$R_{XX}(t_1, t_2) = R_{XX}(t_1 - t_2)$$

Example 2

$$V(t) = \cos(\omega t + \theta)$$

$$E[V(t)] = 0$$

$$E[V^2(t)] = \frac{1}{2}$$

HW Example: White Noise

$$E[X(t)] = 0$$
$$E[X^2(t)] = \sigma^2$$
$$E[X(t)X(t-\tau)] = 0 \quad (\tau \neq 0)$$

Ergodic Process

A stochastic process is **ergodic** if time averages equal ensemble averages.

Time Average

Mean

$$\langle X(t) \rangle = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} x(t)\, dt$$

Autocorrelation

$$R_{XX}(\tau) = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} x(t)x(t+\tau)\, dt$$

Summary

$$\mu_X(t) = E[X(t)]$$
$$R_{XX}(t_1, t_2) = E[X(t_1)X^*(t_2)]$$
$$C_{XX} = R_{XX} - \mu_X(t_1)\mu_X(t_2)$$

SSS $\Rightarrow$ WSS, WSS $\not\Rightarrow$ SSS
Gaussian WSS $\Rightarrow$ SSS
Ergodic: time average = ensemble average